Replicating NGBoost: Probabilistic Regression on Existing and Constructed Data

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Abstract

This project replicates the probabilistic-regression results of natural gradient boosting (NGBoost) on Yacht and Kin8nm, comparing natural-gradient, ordinary-gradient and constant-scale variants. It then evaluates the same variants on five further public datasets, one of them under two partitions, and on synthetic data with known noise scales. Replication was partial: the released code reproduced the natural-gradient advantage on Yacht and the published ordering on Kin8nm, but its Yacht negative log-likelihood (NLL) values were substantially higher than published. A controlled experiment showed that choosing the number of boosting stages by validation NLL instead of squared error brought the two gradient variants' Yacht results close to the published values, while the constant-scale discrepancy remained unresolved. On the new public data, the natural-gradient variant had lower mean NLL than the ordinary-gradient variant in all six settings, and synthetic results favoured input-dependent scales when the noise varied and a constant scale when it did not. A language-model judge of written explanations largely matched one human annotator on individual explanations, but when comparing two explanations it kept the same verdict in only 4 of 15 cases after their order was swapped. The findings support NGBoost in the tested settings and show that implementation choices and automated evaluators need explicit checks.

1 Paper Description and Justification

A house-price forecast of $800,000 does not describe its uncertainty. Two 95% intervals, from $780,000 to $820,000 and from $500,000 to $1.1 million, support different decisions despite sharing that point estimate. Probabilistic regression predicts a distribution over the outcome, conditional on the inputs, so uncertainty can vary between observations. Its fit is evaluated on held-out data.

Duan et al. (2020) introduce Natural Gradient Boosting, or NGBoost, for this purpose.[1] The method extends gradient boosting, which combines a sequence of weak learners, to estimate several parameters of a predictive distribution together. With the Normal distribution used in this project, these parameters describe an input-dependent mean $\mu(x)$ and standard deviation $\sigma(x)$; the implementation represents the latter through $\log \sigma(x)$. Fitting the parameters together can produce poorly balanced updates under an ordinary gradient. For the logarithmic score, the natural gradient rescales the loss gradient using the inverse Fisher information, accounting for how parameter changes alter the distribution (Amari, 1998). The paper's contribution is a boosting framework that combines this update with a chosen distribution, base learner and proper scoring rule.

Original evaluation. The paper compares NGBoost with probabilistic prediction methods including neural-network approaches, Gaussian processes and distributional regression, and separately tests its components through ablations (Duan et al., 2020). Its regression experiments use ten benchmark datasets, Normal predictions and depth-three decision trees. The main measure is mean test negative log-likelihood (NLL), which evaluates the predictive distribution rather than only its mean; lower values are better. For Yacht and Kin8nm, the published protocol repeats random 90/10 training/test holdouts twenty times. Within each training portion, validation log-likelihood selects the number of boosting stages before refitting on the full training portion.

The original paper's ablation table provides the comparison investigated here. On Yacht, the reported mean NLL is 0.20 ± 0.26 for NGBoost, against 3.29 ± 0.20 for ordinary-gradient mul-

[1] https://proceedings.mlr.press/v119/duan20a.html

tiparameter boosting and 2.02 ± 0.21 for the homoscedastic model, whose spread is constant across inputs. The quoted spreads are standard errors over twenty repetitions. On Kin8nm, ordinary-gradient boosting reports -0.52 ± 0.03, NGBoost -0.49 ± 0.02, and the homoscedastic model -0.37 ± 0.05. The ranges for the first two overlap, so the paper's own reporting criterion treats them as tied. These comparisons are the claims this project tests, and they already show that NGBoost does not lead on every dataset. This project examines those three variants on both datasets and extends the comparison to five further public datasets and constructed data with known conditional noise. It does not reproduce the second-order ablation or the full comparison against external model families.

Why this paper. The choice rests on publication quality, reproducibility and a directly testable contribution. NGBoost appeared at the International Conference on Machine Learning in 2020 (Duan et al., 2020). ICML has an A* rating in ICORE2026, as recorded in the project's August 2026 venue check (CORE, 2026). Venue standing is evidence about peer review, and reproducibility is a separate property that this project measures directly.

The official implementation is public under the Apache 2.0 licence (Stanford ML Group, 2020). This project runs that implementation at a recorded commit, with its own scripts for the repair and all experiments.[2] Public data and code make the comparison inspectable. A further criterion was that the released code ran to completion with only minor repairs: two missing import statements, which left the model, splits and stage selection unchanged (Section 3). Running the published implementation, rather than reimplementing it, avoids introducing changes that could alter its results. The project uses scripted data acquisition and preparation with checksums, a synthetic data generator, repeated computational experiments, human annotation with an agreement measure, an LLM-as-judge evaluation and LaTeX reporting. The NGBoost repository's 1,893 stars and 257 forks in October 2026 provide supporting evidence of community interest.

Finally, the ablations connect a specific algorithmic change to distributional performance. Comparing natural and ordinary gradients tests the update rule, while the fixed-spread comparator examines the value of allowing uncertainty to vary with the input. Constructed data makes the second comparison especially informative because the generating spread is known.

2 Original Datasets

The paper's regression benchmark covers Boston, Concrete, Energy, Kin8nm, Naval, Power, Protein, Wine, Yacht and Year MSD (Duan et al., 2020). This replication tests the paper's ablation claim: that natural gradients improve on ordinary gradients and on a constant scale. It uses two datasets on which the published results point in opposite directions. On Yacht Hydrodynamics, a small set of measurements, NGBoost has lower mean NLL than both alternatives. On Kin8nm, a larger simulated set, ordinary-gradient boosting has the lowest published NLL, though within NGBoost's standard-error range. Kin8nm was chosen over Wine because its file comes with the released code and its 8,192 rows are clearly identified; the paper does not say exactly how its Wine data were built. Boston is also unsuitable for new work because of the ethical problem discussed in Section 6.

Yacht Hydrodynamics. The UCI dataset contains 308 observations from experiments involving 22 hull forms at the Delft Ship Hydromechanics Laboratory (Gerritsma et al., 1981).[3] Each row combines hull geometry and a speed condition to predict residuary resistance per unit weight of displacement. Its six predictors are the longitudinal position of the centre of buoyancy, prismatic coefficient, length-displacement ratio, beam-draught ratio, length-beam ratio, and Froude number, a dimensionless measure of speed. The predictors and target are dimensionless. Each of the 22 hull forms appears at 14 speed conditions, accounting for all 308 rows.

Kin8nm. This dataset contains 8,192 observations from a simulation of an eight-link robot arm with revolute joints (DELVE Project, 1996).[4] The prediction target is the distance of the arm's end-effector from a target position. The eight input columns in the released file are named `theta1` through `theta8`, followed by the target column `y`. In the DELVE naming scheme, `8nm` identifies eight inputs, a nonlinear relationship and a medium

[2] Project repository: https://github.com/Sakshamb13579/Replicating-Natural-Gradient-Boosting

[3] https://archive.ics.uci.edu/dataset/243/yacht+hydrodynamics

[4] https://www.cs.toronto.edu/~delve/data/kin/desc.html

Yacht: first two rows, no header

```
-2.3 0.568 4.78 3.99 3.17 0.125 0.11
-2.3 0.568 4.78 3.99 3.17 0.150 0.27
```

Kin8nm: header and first row

```
"theta1","theta2","theta3","theta4",
"theta5","theta6","theta7","theta8","y"
-1.5119208e-02, 3.6074091e-01, 4.6939777e-01,
1.3096745e+00, 9.8802387e-01, -2.5492554e-02,
6.6407094e-01, 6.2762996e-02, 5.3652416e-01
```

Figure 1: Extracts from the original input files. The Kin8nm header and data row are wrapped at commas for display; each occupies one line in the source file. Numeric values are unchanged, and the final field in each data row is the prediction target.

level of output noise. The project uses the CSV distributed with the NGBoost repository, linking the simulation documentation to the actual input used by the released experiment.

Input format. Figure 1 shows extracts from the preserved source files. Yacht is whitespace-separated, has no header and contains seven numeric columns. Kin8nm is comma-separated, has a quoted header and contains nine numeric columns; its values are written in scientific notation. The released script parses these formats separately, then applies the same column rule: the last column is the target and all preceding columns are predictors. It retrieves Yacht at runtime over HTTP from UCI's legacy archive path, while Kin8nm is included in the repository. This project instead uses local copies with recorded checksums, which guard against later changes to those sources, and applies no scaling or imputation. Both datasets have finite values and no missing model inputs. A project-added row identifier tracks splits and saved predictions and is excluded from model features.

3 Replication of Original Work

Implementation and environment. The released-code runs used a CPU-only Python 3.10.0 environment with NGBoost 0.5.11.dev0 installed from commit `6720340` (Stanford ML Group, 2020); Appendix A lists the recorded package versions. The unmodified script loaded the data, then failed with `NameError` at `Dist=eval(args.distn)`: it resolves distribution and score names by evaluating command-line strings, but imports none of them. The three missing names are the two-parameter Normal distribution (`Normal`), a fixed-scale Normal that boosts the mean only (`NormalFixedVar`), and the logarithmic scoring rule (`MLE`, an alias for `LogScore`). Two import statements supplied them in a separate repair worktree, without changing the model, split or selection logic, after which the Yacht and Kin8nm runs completed.[2]

The controlled experiments used the same package versions with single-threaded numerical libraries, removing variation from parallel arithmetic. The historical record lists package versions but no CPU model or thread settings, and these cannot be recovered. Appendix A records both environments. As a guide for rerunning the experiments, the released-code runs took about 40 minutes for Yacht and 45 minutes for Kin8nm across the three variants.

Experimental setup. All variants used the default depth-three tree learner, logarithmic score, learning rate 0.01, full batches and a cap of 2,000 boosting stages. Natural and Ordinary boosted both the mean and log scale of a Normal distribution, with natural gradients enabled and disabled respectively. Homoscedastic boosted only the mean with fixed scale during fitting, then adjusted its predictive scale after refitting, as examined below. The released script generated repeated random 90/10 row holdouts and reserved 20% of each outer-training portion for validation. It chose the number of stages on that validation portion, refitted on the full outer-training portion, and predicted the test rows using the chosen number of stages. Yacht used 20 repetitions and Kin8nm used 5, compared with 20 for each in the original paper.

Three evidence lines. This report distinguishes *published* values from the ablation table of Duan et al. (2020), the historical *released-code run* of the repaired script, and a separate *controlled selector experiment* that applies both selectors to shared fitted trajectories (Table 1). Only the two controlled columns form a within-experiment selector comparison; subtracting a released-code result from a controlled result would mix experiments.

Evaluation measures. Test NLL is the mean negative log predictive density and scores the whole distribution (Gneiting and Raftery, 2007); test RMSE is the root mean squared error of the predicted mean and scores only its centre (Hyndman and Koehler, 2006). Lower is better for both, and NLL can be negative because a continuous density can exceed one. Appendix A gives both definitions, one worked NLL value, and the dispersion conventions of each evidence line. Repeated holdouts share rows, so their summaries should not be read as independent samples.

Dataset	Variant	Published	Released code	Controlled, selected by RMSE	Controlled, selected by NLL
Yacht	Natural	0.20 ± 0.26	2.1060	1.9556	0.4115
	Ordinary	3.29 ± 0.20	11.5473	11.5617	3.5133
	Homoscedastic	2.02 ± 0.21	13.2821	12.7399	12.7399
Kin8nm	Natural	-0.49 ± 0.02	-0.4191	-0.4186	-0.4196
	Ordinary	-0.52 ± 0.03	-0.5078	-0.5083	-0.5075
	Homoscedastic	-0.37 ± 0.05	-0.3623	-0.3685	-0.3685

Table 1: Every numeric result is a mean test NLL; RMSE and NLL in the last two headers name the validation *selector*, not the reported metric. Published values are mean $\pm$ standard error over 20 repetitions from the ablation table (Table 2) of Duan et al. (2020). Both local experiments use 20 repetitions for Yacht and 5 for Kin8nm. Released code denotes the repaired historical run; the two controlled columns use shared trajectories in a separate experiment.

Released-code results. Yacht recovered Natural's advantage: its mean test NLL was 2.1060, although above the published 0.20 (Table 1). Natural beat Ordinary on 19 of 20 holdouts and Homoscedastic on 20 of 20; Ordinary beat Homoscedastic on only 12 of 20. The ranking by mean test NLL therefore reversed the published secondary ordering, with Ordinary ahead of Homoscedastic locally while the original paper places Homoscedastic ahead of Ordinary. That reversal rests on a nearly even fold split, which makes it the weakest of the three local comparisons.

Kin8nm's released-code mean test NLL was lowest for Ordinary at -0.5078, followed by Natural at -0.4191 and Homoscedastic at -0.3623 (Table 1). That ordering matches the original paper's, and it held on all five holdouts for both adjacent pairings. The published Natural and Ordinary standard-error ranges overlap, so the original paper's reporting criterion does not separate those two variants.

Investigating stage selection. The original paper selects stages by validation log-likelihood, but the released script minimises validation squared error while computing and discarding a validation NLL sequence. Squared error and RMSE choose the same stage. To measure the selector's effect, the controlled experiment fixed the outer holdout, the inner 80/20 partition and the model seeds (Appendix A). Both selectors worked from the same fitted sequences, so the only thing that changed between the paired predictions was which stage was chosen. The released script draws its validation split without a fixed random seed and those indices were not saved, so the historical run cannot supply this paired comparison.

For Yacht Natural, NLL selection reduced mean test NLL from 1.9556 to 0.4115 (Table 1), winning on 17 of 20 repetitions. Mean test RMSE rose from 0.6131 to 0.6241, which shows the two measures rewarding different things. Ordinary changed from 11.5617 to 3.5133, approaching the published 3.29. Kin8nm barely moved, with mean NLL changing by -0.0010 for Natural and $+0.0007$ for Ordinary.

Remaining discrepancies. Homoscedastic selected identical stages under validation squared error and validation NLL. In the controlled experiment, the scale at each stage was the root mean square (RMS) of the validation residuals, so ranking stages by validation NLL gave the same order as ranking them by squared error. After refitting, the released script instead sets the predictive standard deviation to the refitted model's predictive variance multiplied by the RMS of the residuals on those same validation rows. By then those rows are part of the refit training set, and multiplying a variance by an RMS does not estimate a residual standard deviation. Both the rows used and the arithmetic therefore depart from the original paper, which sets this comparator's variance to the variance of the training residuals. Together they offer a checkable explanation for the released-code Yacht Homoscedastic NLL of 13.2821 against the published 2.02 (Table 1), although this work does not separate how much each departure contributes. The new-data experiments use a different scale rule, introduced in Section 4.

Under squared-error selection in the controlled experiment, stage selection often reached the 2,000-stage ceiling, on all 20 Yacht Ordinary repetitions and on 7 to 9 of 20 for the other Yacht variants; under NLL selection, Yacht Natural and Ordinary never reached it (Table A1). A separate diagnostic therefore tested whether a higher ceiling would help, comparing 2,000-stage and 4,000-stage limits for Yacht Natural under squared-error selection on shared 4,000-stage trajectories. Mean test NLL

rose from 2.0351 to 2.2029: the larger limit won on 2 holdouts, tied on 5 and lost on 13, so extra headroom made probabilistic performance worse on most of them (Table A2). Homoscedastic still selected the cap under NLL selection (Table A1), so these fixed-budget results do not establish convergence.

Replication outcome. The released-code run reproduced the direction of Natural's advantage on Yacht and the published ordering of mean NLL on Kin8nm, but not Yacht's absolute NLL values. In the controlled selector experiment, NLL selection brought Yacht Natural and Ordinary close to the published means, although Natural remains above its published value (Table 1). The Homoscedastic mismatch is not explained by stage selection; its scale calculation remains a separate unresolved discrepancy.

4 Construction of New Data

Five existing public regression datasets outside the original benchmark test generalisation across domains. Constructed data provides known conditional means and noise scales for directly checking uncertainty estimates.

Public sources and preparation. Table 2 summarises the model inputs and outer partitions. Airfoil Self-Noise contains wind-tunnel measurements predicting scaled sound-pressure level from frequency and four aerodynamic variables (Brooks et al., 1989).[5] Its headerless tab-separated file was assigned descriptive column names. Real Estate Valuation predicts unit-area house price from transaction date, house age, transport proximity, nearby stores and coordinates (Yeh, 2018).[6] The sole spreadsheet worksheet was loaded and its source index removed. Parkinsons Telemonitoring predicts motor severity from voice measurements, age, sex and test time (Tsanas and Little, 2009).[7] Subject identifiers define partitions only; the alternate total-severity outcome is excluded from predictors.

Forest Fires combines weather, spatial coordinates and calendar categories to predict burned area (Cortez and Morais, 2007).[8] Its month and day fields were expanded using fixed calendar categories, giving 29 numeric predictors, and the target became $\log(1+\text{area})$. The transform reduces skew but retains the mass at zero, so a Normal likelihood remains approximate; subsequent metrics are on the transformed scale. Superconductivity uses the 81 numeric predictors in `train.csv` to predict critical temperature (Hamidieh, 2018).[9] The chemical-formula strings in its companion file are not model inputs. All 21,263 rows are used here; an earlier 5,000-row subset remains separate pilot evidence.

Partition units. Row-holdout studies reserve 20% of outer-training rows for validation. Parkinsons instead partitions subjects at both levels, keeping every recording from a subject together; its recording count does not represent independent patients. The additional grouped Airfoil experiment defines 106 conditions by exact tuples of angle, chord, velocity and suction-side displacement thickness, leaving frequency to vary within a condition. The 5 shuffled folds test each condition once, with 20% of outer-training conditions reserved for validation. This tests held-out condition combinations, not necessarily unseen individual feature values. Its partitions and training proportions differ from the row-holdout study, so their results are descriptive comparisons rather than a paired grouping effect.

Constructed regression data. The generator draws independent $x_1, x_2 \sim U(-2, 2)$ and $z \sim N(0, 1)$, then sets

$$\begin{aligned} \mu(x) &= \sin(x_1) + x_2^2, \\ y &= \mu(x) + \sigma(x)z, \\ \sigma_A(x) &= 0.1 + 0.5|x_1|, \\ \sigma_B(x) &= 0.6, \qquad \sigma_C(x) = 2\sigma_A(x). \end{aligned}$$

Regime A provides input-dependent noise, B a constant-noise control, and C greater noise with the same spatial pattern as A. A and B differ in both noise variation and overall variance. The Gaussian generating law tests the chosen likelihood within its assumed distribution family. Each of 10 fresh replicates contains 4,000 base observations, partitioned into 2,400 training, 800 validation and 800 test rows per regime. Inputs, standard-normal draws and partitions are shared across regimes within a replicate, and model seeds are shared across variants and regimes. Only x_1, x_2 enter the models;

[5] https://archive.ics.uci.edu/dataset/291/airfoil+self+noise

[6] https://archive.ics.uci.edu/dataset/477/real+estate+valuation+data+set

[7] https://archive.ics.uci.edu/dataset/189/parkinsons+telemonitoring

[8] https://archive.ics.uci.edu/dataset/162/forest+fires

[9] https://archive.ics.uci.edu/dataset/464/superconductivty+data

Dataset	Rows	Features	Prediction target	Outer partition
Airfoil	1,503	5	Sound-pressure level (dB)	5 row holdouts; separately, 5 condition folds
Real Estate	414	6	Unit-area price	20 repeated row holdouts
Parkinsons	5,875	19	Motor UPDRS score	5 folds over 42 subjects
Forest Fires	517	29	$\log(1 + \text{area})$	20 repeated row holdouts
Superconductivity	21,263	81	Critical temperature (K)	5 repeated row holdouts

Table 2: Additional public datasets after input preparation. Feature counts exclude targets and evaluation identifiers. Row holdouts use 90/10 training/test partitions; grouped folds partition subjects or aerodynamic conditions. Real Estate prices retain the source unit of 10,000 New Taiwan dollars per ping (approximately $3.3\,\mathrm{m}^2$); Forest Fires area is measured in hectares before transformation. UPDRS denotes the Unified Parkinson's Disease Rating Scale.

Forest Fires: selected fields, zero-based row 138

```
Source: month=jul, day=tue, area=0.36
Prepared: month_jul=1, day_tue=1
log1p_area=0.3074846997479606
```

Synthetic: replicate 1, row 0, training

```
x1=0.047286498801026866
x2=0.7557451779003403
true_mu=0.6184196524866178
noise_z=-1.6109632612127667
```

Regime	True scale	Target y
A	0.12364324940051344	0.4192349202054232
B	0.6	−0.3481583042410422
C	0.24728649880102688	0.2200501879242286

Figure 2: Selected fields from preserved data, with numeric precision retained. Forest Fires shows a source-to-input transformation; the synthetic rows share predictors, mean and noise draw while their scales and targets differ. True mean, true scale and noise draw are evaluation-only; model inputs are x_1, x_2.

true means, scales and noise draws are evaluation-only. An earlier constructed-data study used one draw rather than independent replicates, so its results are not pooled with the ten-replicate summaries reported here. Figure 2 shows genuine prepared values; Appendix B gives seed provenance and further examples.

Quality and evaluation protocol. Preparation checks finite model values, excludes identifiers from features, freezes split indices and records source and processed-file checksums. No scaling, imputation or outcome-based row filtering is applied, and source duplicates are retained. Full Superconductivity contains 66 exact duplicate extra rows and 6,093 repeated-feature extra rows. Among the distinct feature tuples, 2,374 carry more than one target value, spanning 8,423 rows. Such tuples can cross row partitions, so the results do not show generalisation to unseen materials. New-data runs retain the learner, score, learning rate and 2,000-stage budget from Section 3, select stages by validation NLL, and refit on training plus validation rows. For Homoscedastic, each stage's scale is the RMS of the validation residuals and the selected scale is retained after refitting, unlike the released-code calculation in Section 3. This comparator uses validation outcomes for both scale estimation and stage selection; the other variants use them only for stage selection. Regression targets come from the public sources or generator; only the authored explanations evaluated in Section 5 receive human labels.

5 Results on New Data

5.1 Predictive Performance

The three variants were evaluated on the additional public datasets and synthetic regimes introduced in Section 4. The learner and training budget remain unchanged from the replication. Stage selection follows the controlled replication's validation-NLL rule rather than the released script's squared-error rule. Homoscedastic uses the revised scale calculation described in Section 4. Parkinsons and grouped Airfoil additionally test predictions for unseen subjects or conditions.

Public-data findings. Natural has lower mean test NLL than Ordinary in all 6 settings, agreeing with Yacht's direction but reversing the Kin8nm ordering in both the published and local replication results (Tables 3 and 1). Natural achieves the lowest NLL on Parkinsons, Real Estate and full Superconductivity, while Homoscedastic performs best on both Airfoil evaluations. Forest Fires shows little numerical separation between the variants. NLL and RMSE give different rankings in four of the six settings, agreeing only on the Airfoil evaluations: accurate point predictions do not necessarily identify the best predictive distributions.

Interpreting coverage. On Superconductivity, Ordinary's highest coverage comes with mean interval widths of 115.79 against 34.55 for Natural,

Setting	Natural NLL	Natural RMSE	Natural C_{95}	Ordinary NLL	Ordinary RMSE	Ordinary C_{95}	Homoscedastic NLL	Homoscedastic RMSE	Homoscedastic C_{95}
Airfoil, rows	2.2470	2.592	0.913	2.9465	5.414	0.956	**2.1567**	2.073	0.957
Airfoil, conditions	2.5453	3.441	0.938	3.1586	5.928	0.917	**2.4559**	2.719	0.956
Forest Fires	**1.7785**	1.410	0.935	1.7826	1.404	0.933	1.7787	1.415	0.935
Parkinsons, subjects	**3.6301**	8.572	0.919	3.6486	8.358	0.924	3.6662	9.000	0.932
Real Estate	**3.6441**	8.059	0.959	4.0367	13.506	0.971	3.7928	7.955	0.959
Superconductivity, full	**3.5158**	11.558	0.945	4.7108	33.858	0.990	3.7974	10.762	0.942

Table 3: Additional public-data results: unweighted means of fold or repetition metrics under the partitions in Table 2. Lower NLL and RMSE are better; C_{95} is the fraction of test outcomes inside the nominal 95% prediction interval, with 0.95 the nominal target rather than a score to maximise. Bold identifies the lowest mean NLL within each setting, not a significance claim. RMSE retains each dataset's target units; Forest Fires uses the transformed target. NLL magnitudes are not comparable across target scales.

and with worse NLL and point predictions, so coverage alone does not indicate better probabilistic performance (Table 3; Appendix C).

Generalisation and training budget. All three variants have higher mean NLL when Airfoil conditions are held out together than under row holdouts (Table 3). This tests prediction for unseen condition combinations, although changed training proportions mean that the difference cannot be attributed to grouping alone. For Superconductivity, repeated predictor combinations can cross partitions, so the results concern new rows rather than necessarily new materials. Moreover, 12 of its 15 fits select the maximum stage count: these are fixed-budget results, not evidence that further training cannot help. Appendix C records aggregation details and cap counts.

Constructed-data findings. Natural has the lowest NLL in the input-dependent-noise regimes A and C, while Homoscedastic leads in the constant-noise control B (Table 4). The paired replicate comparisons support these orderings and extend Natural's Yacht advantage over Ordinary across all three regimes (Appendix C). Homoscedastic has the highest NLL in the local Yacht and Kin8nm replication runs, but the lowest in B. Its revised scale calculation means that this reversal cannot be attributed to the dataset alone.

The scale errors help explain this contrast: Natural recovers the varying noise more accurately in A and C, whereas a single scale is more accurate when the noise is constant in B (Table 4). Natural and Homoscedastic nevertheless have nearly identical RMSE and overall 95% coverage in A and C, for example 0.6820 against 0.6821 and 0.935 against 0.934 in A (Appendix C). Because the true noise scales are known, the synthetic data show that models with similar point predictions and coverage can spread uncertainty differently across inputs. This supports input-dependent scale estimation in the varying-noise settings tested here. These findings concern simulated outcomes that are Normal given the inputs. Regimes A and B differ in average noise variance as well as noise structure, so their comparison does not isolate noise structure alone.

Regime	Variant	NLL	Scale MAE	Oracle gap
A	Natural	**0.8249**	0.063	0.077
	Ordinary	0.8563	0.089	0.109
	Homoscedastic	1.0379	0.259	0.290
B	Natural	0.9456	0.043	0.032
	Ordinary	0.9916	0.089	0.078
	Homoscedastic	**0.9364**	0.012	0.023
C	Natural	**1.4876**	0.107	0.047
	Ordinary	1.5164	0.136	0.076
	Homoscedastic	1.7220	0.516	0.281

Table 4: Means over 10 fresh synthetic replicates per regime. Scale MAE compares predicted standard deviation with the true conditional noise scale. Oracle gap is fitted NLL minus NLL under the true generating distribution on the same test outcomes; it is a reference difference, not a guaranteed nonnegative bound. Bold marks the lowest mean NLL per regime.

5.2 LLM-as-Judge Evaluation

A separate LLM-as-judge experiment evaluates explanation clarity, interval interpretation and evidence support, complementing numerical assessment of predictive distributions. Zheng et al. (2023) study language models as evaluators. The judge assesses written explanations, not regression variants.

Evaluation design. MiniCPM4.1-8B (OpenBMB, 2025), using Q8_0 quantisation and served locally through Ollama 0.9.6 (Ollama, 2025), evaluated 15 held-out Natural predictions,

Criterion	Agreement	κ	Baseline
Clarity	90.0%	0.000	90.0%
Interval consistency	86.7%	0.636	83.3%
Evidence support	93.3%	0.867	50.0%

Table 5: Pointwise judge agreement with one human annotator over 30 explanations per criterion. Baseline always predicts that criterion's majority human label; κ is Cohen's chance-adjusted agreement (Cohen, 1960).

with 3 cases each from synthetic A, B and C, grouped Airfoil and Real Estate. Each case supplied a prediction packet, one intended-faithful description and one controlled edit. Across the 15 cases, five edits targeted interval misinterpretation, five unsupported causes and five obscuring jargon. Packets omitted observed outcomes; synthetic packets additionally supplied the known generating law. Implementation checks and a three-case smoke test preceded evaluation; the same prompt was then held fixed, with temperature 0 and seed 0, to measure the judge's behaviour. All 60 requests produced valid structured outputs: 30 pointwise assessments on three binary criteria and 30 pairwise comparisons covering both answer orders. Clarity assesses understandable wording, interval consistency assesses numbers, units and statistical interpretation, and evidence support assesses whether substantive claims follow from the packet. A human annotator assessed all 30 explanations using the supplied evidence and rubric (Appendix C). The presented items contained neither judge verdicts nor the authored edit categories. These annotations are the agreement reference; the edit categories are not reference labels.

Agreement and failure modes. Evidence support agrees most strongly with the human reference, while clarity's 90% agreement equals its majority baseline because the judge passes every explanation (Table 5). All disagreements on interval consistency and evidence support are explanations that the annotator accepted and the judge failed (Appendix C). On three jargon-heavy explanations, the annotator identifies a clarity problem while accepting the interval statements; the judge assigns the opposite labels. This suggests that the judge does not consistently separate readability from statistical correctness. Annotator clarity labels also vary across the five items sharing the same final jargon sentence.

Answer-order sensitivity. Only 4 of 15 cases retain the same underlying pairwise verdict after the explanations are swapped, and the judge favours the second position overall (Appendix C), consistent with the type of position bias documented by Zheng et al. (2023). In all five interval-misinterpretation cases, it prefers the explanation that confuses a prediction interval with a confidence interval and claims a guarantee for the next outcome, whenever that explanation appears second. Yet the same judge rejects all five texts on interval consistency when assessing them individually. Pointwise agreement therefore does not establish reliable pairwise explanation selection.

Agreement was measured against one human annotator and describes this evaluation set; variation between annotators was not assessed. Because the examples share five explanation templates, the results describe the tested examples rather than broad performance across explanation styles.

6 Ethical Considerations

Publication venues increasingly expect an ethics statement; Benotti and Blackburn (2022) classify such sections by whether they address research benefits, potential harms and affected vulnerable groups, and Karamolegkou et al. (2025) catalogue the concerns raised across ACL papers. This section addresses all three, for the original work and for the new data.

Responsible use of uncertainty. The original NGBoost work makes predictive uncertainty available alongside point estimates (Duan et al., 2020), potentially supporting better-informed decisions. However, a predictive distribution can create misplaced confidence when its assumptions fail or its intervals are interpreted as guarantees. The Superconductivity results illustrate why coverage alone is insufficient: high coverage accompanied much wider intervals and worse point predictions (Section 5). In health or financial applications, misleading uncertainty could affect treatment or valuation decisions. Neither the original benchmark results nor this extension establishes suitability for such deployment; calibration and usefulness require assessment in the intended setting.

Benchmark data in the original work. The original regression benchmark includes the Boston housing dataset (Section 2). That dataset carries a constructed variable derived from the proportion of Black residents in each town, built on an assumption that racial segregation raised house prices; the

scikit-learn maintainers removed its loader for this reason (scikit-learn developers, 2021). The original paper reports an averaged benchmark result over that dataset without discussing how it was built. A paper published on this data today would be expected to state such provenance; the Boston data are not used in this project.

Licensing, ownership and attribution. Open access supports replication without transferring ownership of reused material. NGBoost's Apache 2.0 licence requires distributed copies to retain relevant notices and the licence, and modified upstream files to identify changes (Stanford ML Group, 2020). UCI lists the five additional public datasets (Section 4) under CC BY 4.0, which requires credit, a licence link and an indication of changes when the data or adaptations are redistributed (Creative Commons, 2013). Section 4 identifies their creators and preparation steps. Yacht is likewise listed under CC BY 4.0, while no separate licence statement was located for Kin8nm, which reached this project through the NGBoost repository; it is therefore cited rather than redistributed. The judge uses a third-party quantisation of the Apache 2.0 MiniCPM4.1-8B release (OpenBMB, 2025), referenced by repository and checksum rather than redistributed (Appendix C). The public repository released with this report (Section 1) will separate upstream material from project-authored files and keep third-party terms on the upstream files rather than placing everything under one project licence. These practices follow three principles of the Australian Code for the Responsible Conduct of Research: fairness, which includes citing and crediting the work of others; transparency in reporting methods, data and findings; and accountability (National Health and Medical Research Council et al., 2018).

Data sensitivity and generalisation. Parkinsons contains health-related measurements, while Real Estate includes transaction locations. Permission to reuse data does not remove privacy considerations or establish consent for every downstream application. Parkinsons subject identifiers define disjoint partitions rather than predictors; this reduces evaluation leakage but does not itself establish privacy protection. Aggregate results do not demonstrate equal performance across demographic groups, populations or locations, and subgroup fairness was not evaluated. The synthetic experiments use no personal records, and because their generating distributions are known, the predicted uncertainty can be checked against the truth. Those outcomes are Normal given the inputs, with a specified mean and scale, so the synthetic results do not establish how the method would behave on real clinical or property data.

Transparent reporting and automated judgement. Preserved outputs, fixed protocols, seeds and checksums make the reported comparisons open to scrutiny. Reporting the published results, the released-code run and the controlled selector experiment as separate evidence lines avoids implying stronger replication evidence than was obtained. The explanation judge rejected all five texts that misdescribe the prediction interval when scoring them individually, yet preferred those texts whenever they appeared second in a side-by-side comparison (Section 5). Its agreement scores therefore do not justify autonomous explanation selection, and anyone using it should keep human review of statistical interpretation and evidential support.

7 Conclusions and Reflections

Extent of replication. Replication was partial (Section 3). Once two missing import statements were added, the released-code run recovered Natural's advantage on Yacht and the published ordering on Kin8nm, but its Yacht NLL values were substantially higher than published. The controlled selector experiment showed that choosing stages by validation NLL rather than squared error brings Yacht Natural and Ordinary close to the published values, while the Homoscedastic discrepancy remains unresolved.

Generalisation to new data. Generalisation was qualified. Natural had lower mean NLL than Ordinary in all six additional public-data settings, but Homoscedastic led on both Airfoil evaluations and Forest Fires showed little separation. On synthetic data, Natural estimated the noise scale more accurately and had the lowest NLL when the noise varied with the inputs, while Homoscedastic did both when the noise was constant. In the varying-noise regimes the two had nearly identical RMSE and overall coverage, so those measures alone would not have shown how differently they spread uncertainty across inputs. These conclusions are limited to the tested partitions, synthetic distributions and training budget. They rest on validation-NLL stage selection and the revised Homoscedastic scale rule rather than the released script's rules.

Methodological reflections. The stage-

selection rule, an implementation detail outside the learner, changed the reported results substantially; the fixed-scale calculation is a further candidate whose effect was not isolated. In constructing the new data, holding out whole subjects or conditions tested prediction for subjects and condition combinations absent from training, and known synthetic noise scales let predicted uncertainty be checked against the truth. In hindsight, regimes A and B should have had equal average noise variance, so that their comparison isolated noise structure.

The judge experiment showed that an evaluator must itself be tested. It largely agreed with the annotator on individual explanations, yet when comparing two explanations it kept the same verdict in only 4 of 15 cases after their order was swapped, and it preferred interval misinterpretations it had rejected individually whenever they appeared second. Annotation also exposed design limits. The annotator's clarity labels differed across items sharing the same jargon sentence; because the rest of each text differs, this is not necessarily inconsistency, but a sharper definition of clarity would make such labels easier to interpret. A future explanation set should also use more than one phrasing per edit type, to test whether the findings extend beyond repeated templates.

Overall, the experiments support NGBoost for probabilistic regression in the tested settings, provided its uncertainty is checked rather than taken on trust. Because an interval can be read as a guarantee and a fluent explanation can make a wrong interval statement persuasive, responsible use also requires calibration in the intended setting and human review of explanations before they inform decisions (Section 6). More broadly, the model, the data used to test it and the tools used to judge its explanations each need checking in their own right.

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A Replication Environment and Diagnostics

Recorded environments. The historical environment record lists Python 3.10.0, NGBoost 0.5.11.dev0 installed in editable mode from commit `6720340`, NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3 and scikit-learn 1.6.1. It does not record a CPU model or numerical-library thread settings. The controlled selector experiment records the same package versions on Linux 6.8.0-139-generic, x86-64, with glibc 2.39. Before execution, `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, `MKL_NUM_THREADS` and `NUMEXPR_NUM_THREADS` were each set to 1. These restrictions support reproducibility within the recorded environment; they do not establish identical arithmetic across platforms.

Controlled split provenance. For zero-based repetition index k, the controlled selector experiment uses inner-split seed $10000+k$, independently of model fitting, seeds stage-selection models with $20000 + k$ and refit models with $30000 + k$, and preserves the resulting indices. The separate Yacht Natural headroom diagnostic also freezes inner partitions with seed $10000 + k$, seeds inner models with $20000+k$ and refit models with $30000+k$, and compares both caps on shared 4,000-stage fits. Its NLL means are diagnostic results, not substitutes for the controlled selector experiment's means.

Measures and dispersion. For observations y_i, predictive densities p_i and predicted means $\hat{\mu}_i$,

$$\mathrm{NLL} = -\frac{1}{n}\sum_{i=1}^{n}\log p_i(y_i),$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{\mu}_i)^2}.$$

As a worked value, a Normal prediction with $y = \mu = 0$ and $\sigma = 0.1$ has NLL $\log(0.1\sqrt{2\pi}) \approx -1.384$. The original paper reports standard errors over its repetitions, whereas the released script prints population standard deviations. Table 1 therefore carries the published standard errors only; adding local spread would make that cross-protocol table too wide to read, and the per-repetition values are kept in the run records.

B New-data Provenance and Examples

Frozen preparation. Public-data audits retain the UCI dataset records, source citations, downloaded-file hashes and column roles. Prepared public-data CSV files preserve source row order; the historical Superconductivity pilot selects a subset and sorts its sampled source indices before storage. The full-data experiment retains all source rows and is not pooled with that pilot. Forest Fires retains four exact duplicate rows and allows repeated spatial cells across row partitions. Real Estate row partitions mix transaction dates and may share locations, so they do not represent future-period or location-disjoint prediction.

Dataset	Variant	Selector RMSE	Selector NLL
Yacht	Natural	9/20	0/20
	Ordinary	20/20	0/20
	Homoscedastic	7/20	7/20
Kin8nm	Natural	2/5	0/5
	Ordinary	2/5	2/5
	Homoscedastic	1/5	1/5

Table A1: Repetitions selecting the 2,000-stage cap in the controlled selector experiment, out of all repetitions for that dataset.

Yacht Natural headroom diagnostic	Value
Mean test NLL, cap 2,000	2.0351
Mean test NLL, cap 4,000	2.2029
Mean paired NLL change	+0.1678
Cap 4,000 wins / ties / losses	2 / 5 / 13

Table A2: Separate controlled cap diagnostic over 20 Yacht Natural repetitions, using validation RMSE selection on shared 4,000-stage trajectories. Wins denote lower test NLL for the larger cap.

Randomness and partitions. Repeated public-data row holdouts use sequential permutations from NumPy's legacy random generator (`RandomState`), seeded with 1, with inner-split seed $10000 + k$ for zero-based repetition or fold index k, counted separately in each experiment. Grouped Airfoil and Parkinsons use 5 shuffled partitions of sorted condition or subject identifiers with seed 1; inner validation holds out 20% of outer-training groups with seed $10000 + k$. All variants reuse the saved indices, with stage-model seed $20000 + k$ and refit-model seed $30000+k$. For fresh synthetic replicate $r \in \{1, \ldots, 10\}$, PCG64 generators use data seed r and split seed $1000 + r$. A single permutation assigns the training, validation and test labels, shared across regimes. Stage-model and refit-model seeds are $20000 + 100r$ and $30000 + 100r$, respectively. Each synthetic manifest records its generator hash, data seed, split seed and generator type.

Additional prepared-data snippets. The fol-

lowing extracts show selected fields from the first prepared row of each dataset, without rounding. They are field projections rather than complete input records. Row and group identifiers are excluded from predictors; the prediction target is named explicitly in each extract.

Airfoil, row 0

```
condition_id=23
frequency_hz=800.0
angle_deg=0.0
chord_m=0.3048
velocity_ms=71.3
scaled_sound_pressure_db=126.201
```

Real Estate, row 0

```
X1 transaction date=2012.9166667
X2 house age=32.0
X5 latitude=24.98298
X6 longitude=121.54024
Y=37.9
```

Parkinsons, row 0

```
subject#=1
age=72
sex=0
test_time=5.6431
motor_UPDRS=28.199
```

Superconductivity, row 0

```
number_of_elements=4
mean_atomic_mass=88.9444675
critical_temp=29.0
```

Airfoil's target is `scaled_sound_pressure_db`; Real Estate's is `Y`; Parkinsons uses `motor_UPDRS`; Superconductivity uses `critical_temp`. Airfoil's displacement-thickness predictor and the unshown predictors in the other extracts remain model inputs.

C Additional New-data Results

Aggregation and interval width. All public-data summaries average the saved fold or repetition metrics without weighting by fold size. For Parkinsons, each fold metric averages recordings, so patients with more recordings contribute more within their fold. Table C1 supplements the main results with mean prediction-interval widths. Repeated row holdouts share observations, so their spread describes variation across partitions rather than independent-sample uncertainty. The two Airfoil settings and the Superconductivity pilot and full-data runs are not paired comparisons. The 5,000-row pilot is not pooled with the full-data experiment because they evaluate different training and test samples. On full Superconductivity, the stage cap is selected in 3 of 5 Natural fits, 4 of 5 Ordinary fits and all 5 Homoscedastic fits, giving 12 of 15 overall.

Setting	Natural	Ordinary	Homosc.
Airfoil, rows	7.67	16.74	8.47
Airfoil, conditions	11.30	18.70	11.89
Forest Fires	5.35	5.38	5.36
Parkinsons, subjects	30.31	30.32	32.69
Real Estate	29.74	47.48	31.84
Superconductivity, full	34.55	115.79	43.47

Table C1: Mean nominal 95% prediction-interval widths, in each dataset's target units, including the transformed Forest Fires scale.

Regime	Variant	NLL SD	RMSE	C_{95}	Width
A	Natural	0.0348	0.6820	0.935	2.271
	Ordinary	0.0412	0.7029	0.924	2.302
	Homosc.	0.0324	0.6821	0.934	2.672
B	Natural	0.0242	0.6181	0.943	2.392
	Ordinary	0.0294	0.6356	0.924	2.347
	Homosc.	0.0249	0.6167	0.948	2.391
C	Natural	0.0339	1.3512	0.937	4.503
	Ordinary	0.0335	1.3936	0.927	4.592
	Homosc.	0.0318	1.3520	0.936	5.304

Table C2: Additional synthetic results over 10 replicates per regime. Except for NLL SD, entries are means of test metrics; coverage and width concern nominal 95% prediction intervals.

Synthetic dispersion and aggregate coverage. The 10 fresh synthetic replicates use independent base data and split draws, with regimes paired within each replicate. Table C2 reports sample standard deviations of replicate NLL and complements the main scale-recovery results with RMSE, coverage and width. The mean oracle NLL references are 0.7477 in A, 0.9137 in B and 1.4408 in C. They evaluate the true conditional mean and scale on the same finite test outcomes as each fitted model. The earlier single-draw synthetic experiment is excluded from these summaries.

Paired synthetic comparisons. Table C3 counts strict NLL wins on the shared test outcomes within each replicate. Natural beats Ordinary in all 10 replicates of every regime and Homoscedastic in every replicate of A and C. Homoscedastic beats Natural in 9 of 10 B replicates.

Pointwise disagreements. Six explanations differ between the judge and annotator on at least one criterion. Table C4 gives the direction of each criterion's disagreements. Both evidence-support disagreements concern intended-faithful Real Estate descriptions. Three jargon explanations account for all clarity disagreements and three of the four interval disagreements: the annotator rejects clarity but

Comparison	A	B	C
Natural beats Ordinary	10/10	10/10	10/10
Natural beats Homosc.	10/10	1/10	10/10
Homosc. beats Natural	0/10	9/10	0/10

Table C3: Paired counts of lower test NLL across 10 replicates per regime.

Criterion	Judge pass, annotator fail	Judge fail, annotator pass
Clarity	3	0
Interval consistency	0	4
Evidence support	0	2

Table C4: Directional disagreement counts out of 30 explanations per criterion. These are comparisons with the observed annotation, not adjudicated error counts.

accepts interval consistency, while the judge does the reverse. The annotator accepts clarity on two other jargon explanations, despite all five sharing the same final sentence.

Pairwise order checks. Table C5 distinguishes answer-identity consistency across swaps from literal position choices. The five interval-misinterpretation edits all fail interval consistency individually but win the pairwise comparison when placed second. These contrasts were checked against the raw responses using the frozen verdict parser, not inferred from the edit categories alone.

Annotation procedure. One annotator labelled all 30 explanations on the three criteria, using the rubric text supplied to the judge. The annotation form contained an evidence packet, an explanation and fields for the three binary labels and a comment. Items were presented individually, and the annotator's responses were recorded in a single sheet. That sheet holds only an opaque item identifier, the evidence packet, the explanation, the three binary labels, an annotator identifier and a free-text note. It carries no construction intent, no case identifier and no judge output. Shuffling did not keep related items apart: 4 of the 15 explanation pairs are adjacent. All 90 labels and all 30 notes are complete, and the evidence and explanation fields are unchanged from the archived blank sheet. The completed sheet and its SHA-256 are preserved with the batch.

Judge reproducibility. The judge weights are a third-party Q8_0 quantisation of MiniCPM4.1-8B (OpenBMB, 2025), identified by the Hugging Face repository, revision and file hash below.

Check	Count
Same underlying verdict after swap	4/15
Changed verdict after swap	11/15
First position preferred	4/30
Second position preferred	20/30
Explicit tie	6/30
Interval edit rejected individually	5/5
Interval edit preferred when second	5/5

Table C5: Pairwise consistency counts use 15 evidence cases; position counts use 30 displayed orders. The final two rows concern the same five interval-error explanations.

```
base    openbmb/MiniCPM4.1-8B
gguf    mradermacher/MiniCPM4.1-8B-GGUF
rev     7032ea8d4fc677dd3263c25d7c5343a8094c3151
sha256  9e939664e74b400b618d0e700a0794a2
        d58a889c487ec6c62983354b407bce7e
```

Ollama 0.9.6 (Ollama, 2025) served sequential CPU requests with a 2,048-token context, 4 threads and at most 160 output tokens. The ChatML prefix used non-reasoning mode, and JSON schemas constrained the verdict format; outputs are discrete labels rather than calibrated label probabilities. The three-case development smoke test exposed order sensitivity; its prompt was retained unchanged for evaluation rather than retuned against evaluation results. Raw prompts, responses, the frozen code and model configuration are preserved with checksums.